

Skills Summary

Linear Contexts, Tables, Equations, and Graphs

Use this reference while completing your worksheet or when studying.

One relationship, four faces. Whatever form you are handed, you are always hunting the same two numbers: the **rate of change** and the **initial value**.

1. Reading values and changes off a table or graph

From a table: each column is one (x, y) pair. A relationship is linear when equal steps in x produce equal steps in y — check that before you trust any prediction.

From a graph: only read points where the line crosses a grid *intersection*; an estimated point produces a wrong slope that looks plausible.

Change vs. value: “what is y ” asks for one entry; “how much did y change” asks for a subtraction between two entries.

Example: A table gives x : 0, 2, 4, 6 and y : 5, 11, 17, 23. How much does y change from $x = 0$ to $x = 4$?

Step 1. The question asks for a change, not a value, so read two entries and subtract instead of hunting for a single cell.

Step 2. At $x = 4$ the table gives $y = 17$; at $x = 0$ it gives $y = 5$. The change is $17 - 5$. change in $y = 12$

Try it: A table gives x : 0, 3, 6, 9 and y : 40, 34, 28, 22. How much does y change from $x = 0$ to $x = 6$?

check: $12 - 28 = 40$

2. Rate of change from any two points

Rule: for any two points (x_1, y_1) and (x_2, y_2) of a linear relationship,

$$\text{rate of change} = m = \frac{y_2 - y_1}{x_2 - x_1}$$

Sign check first: rising line, positive rate; falling line, negative rate. Do this before dividing — it catches a swapped subtraction instantly.

Keep both differences in the same order: top y_2 with bottom x_2 .

Example: A line passes through $(1, 4)$ and $(5, 16)$. Find its rate of change.

Step 1. Two points of one linear relationship are all the slope formula needs, so no equation or graph is required here.

Step 2. $\frac{16 - 4}{5 - 1} = \frac{12}{4}$, and the line rises, so a positive answer is expected. $m = 3$

Try it: A line passes through $(2, 9)$ and $(6, 1)$. Find its rate of change.

check: $2 - 9 = 1$

3. Building the equation and using it both ways

Rule: $y = mx + b$, where m is the rate of change and b is the initial value (the y when x is zero).

In a story: a one-time amount (joining fee, starting height) is b ; an “each” or “per” amount is m .

Forwards — given x , substitute and evaluate.

Backwards — given y , set the expression equal to it and solve for x .

If the table has no $x = 0$ row, work back one step at a time to reach it.

Example: A pool holds water 30 cm deep and is filling at 4 cm per hour. How deep is it after 7 hours?

Step 1. The depth it already has is the initial value and the “per hour” amount is the rate, so the story hands over both numbers directly.

Step 2. The equation is $h = 4t + 30$; substituting $t = 7$ gives $4(7) + 30 = 28 + 30$. $h = 58$

Try it: A second pool holds water 25 cm deep and fills at 6 cm per hour. How deep is it after 5 hours?

check: 58

(!) Watch out: when a table steps by more than one, the change in y is *not* the rate of change — you must still divide by the change in x . And a rate read off a falling line must come out negative.
